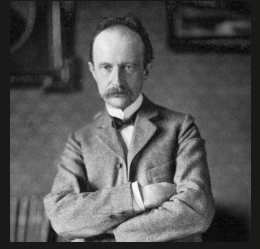


The Quantum Physics Timeline

1900

Max Planck correctly derives the formula to describe blackbody radiation, assuming that atoms can only transfer energy in discrete bundles, or quanta. Energy quanta are proportional to the atomic oscillation frequency, the proportionality factor is now called the Planck's constant.



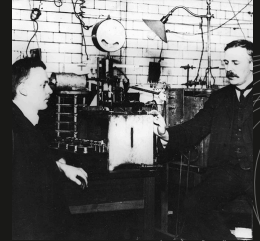
1905

Albert Einstein explains why the photoelectric effect depends on the frequency of light. Light carries energy in quanta (photons) that are proportional to the frequency of light and to Planck's constant. Thus electrons can absorb energy only if the incident light is above a certain frequency.



1909

Marsden, Geiger and Rutherford bombard gold atoms with heavy particles to find out what is the atomic internal structure. They discover that atoms are made of a dense core, with electrons orbiting around like in the solar system. In such a model, electrons should immediately collapse on the nucleus according to classical physics.



1913

Niels Bohr solves the atom instability problem, by assuming that electrons can only radiate or absorb energy in quanta. This explains why atoms are stable and also explains the observation of atomic spectral lines (light is emitted only at specific frequencies when electrons change their orbit).



1923

Louis de Broglie introduces the idea that everything should obey the Einstein formula for the momentum of a photon. This is the particle-wave duality principle. Not only should light behave like a particle, but particles should also behave like waves and undergo phenomena like diffraction. Electron diffraction is observed 4 years later by Davisson and Germer.



1925

Heisenberg (matrix mechanics) and Schrodinger (wave mechanics) break away from astronomical orbits pursued by Bohr and develop a radically new approach to the description of electrons behaviour in atoms. This is the beginning of modern quantum mechanics and the introduction of the probabilistic approach.



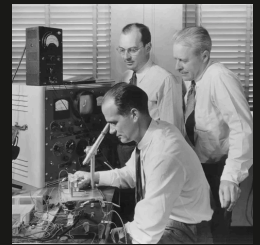
1927

Heisenberg enunciates his Uncertainty Principle. It is impossible to know both the position and momentum of a particle with infinite precision, even if you have a perfect instrument capable of making perfect measurements. Uncertainty is a basic part of physics, at the quantum level.



1947

The transistor is invented at Bell Labs. The transistor is one of the most important technological applications of quantum physics, and made it possible the development of modern electronics (from computers and mobile phones to LEDs and solar cells).



1982

Feynman argues that classical computers are inefficient for simulating quantum systems because the resources required grow exponentially with system size. He proposes the idea of quantum computers that utilize quantum mechanics to simulate physics more efficiently. It marks the beginning of the research on quantum computing.

